

Charotar University of Science and Technology
Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/ MICROBIOLOGY/BIOCHEMISTRY/)
SEMESTER I EXAMINATION(BACKLOG)
BT/MI/BC 704: MOLECULAR BIOLOGY

Date: 29/04/2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in Part A should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of Part B in separate answer sheets.

Total marks: 20

PART A

Q1. Choose the correct option and put $\sqrt$ mark in front of it:

1. DNA is said to be negatively supercoiled when
 - a). $\Delta L_k < 0$
 - b). $\Delta L_k > 0$
 - c). $\Delta L_k = 0$
 - d). none of the above
2. The number of introns per gene in yeast, fruit fly and humans is in the following order
 - a). Fruit fly > human > yeast
 - b). yeast < fruit fly < human
 - c). fruit fly < yeast < human
 - d). human < yeast < fruit fly
3. The super coiled and relaxed DNA can be distinguished by binding of the DNA molecule with the following compound
 - a). Congo red
 - b). acridine orange
 - c). psoralen
 - d). histone H1
4. A structure called "tetraloop" is present in the following macromolecule
 - a). DNA
 - b). RNA
 - c). protein
 - d). lipid
5. The reaction catalyzed by DNA photolyase is an example of:
 - a) Excision repair
 - b) Direct repair
 - c) recombination repair
 - d) None of above
6. Which of the following statements about DNA replication in prokaryotes is false?
 - a). DNA synthesis proceeds along both strands at the same time.
 - b). DNA synthesis proceeds in both directions from an origin simultaneously.
 - c). Leading strand synthesis occurs in short bursts, creating a series of fragments.
 - d). A polymerase complex can be found at the replication fork.
7. In E. coli DNA replication, primer is:
 - a). A deoxyribonucleotide short sequence
 - b). A short RNA annealing to the 3' end of the template strand
 - c). A short RNA complementary to the 5' end of the leading strand
 - d). Synthesized by DNA polymerase I
8. Telomeres can best be characterized by the following statement: _____.
 - a). found at the ends of linear chromosomes
 - b). composed of short tandem repeats
 - c). highly conserved in evolution
 - d). All these statements characterize telomeres.

9. RNA polymerase II transcribes

- a). mRNA b). rRNA c). tRNA d). None of the above

10. RNA editing utilise

- a). gRNA b). rRNA c). tRNA^d. d). miRNA

11. TBF is a subunit of

- a). TFIIA b). TFIID c). TFIIB d). None of the above

12. Bacterial RNA polymerase consists of multiple subunits

- a). [True] b). [False]

13. The stage of translation in which an aminoacyl-tRNA binds to AUG on the A site of the ribosome is

- a) Initiation b) Elongation
c) Termination d) None of the above

14. A codon may specify for more than one amino acid.

- a). [True] b). [False]

15. Which of the following is true for the genetic code?

- a). Genetic code is overlapping. b). There is punctuation between codons.
c). Genetic codon follow wobble hypothesis. d). All of the above

16. The scientist Wobble proposed wobble hypothesis.

- a). [True] b). [False]

17. The product of *lacI* is

- a) Repressor b) promoter c) activator d) none of the above

18. Which of the following can physically block interactions between promoter DNA sequences and the proteins needed to initiate transcription?

- a) RNA b) nucleosome c) histone proteins d) HO gene

19. Methylation of both DNA and histones is a feature of active chromatin.

- a). [True] b). [False]

20. HAT stands for

- a). Histone acetyl transferase b). Histone aminoacyl transferase
c). Histone acetate transferase d). Histone aryl transferase

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Part B

Total Marks: 50

Section I

Q1. (A). Discuss **any one** of the following:

(4)

- i. Structure of tRNA
- ii. Packaging of eukaryotic genomes in nucleosomes
- iii. Structure of DNA

Q1. (B). Write a note on **any one** of the following:

(3)

- i. Centromere
- ii. Reassociation kinetics
- iii. Structure of eukaryotic mRNA

Q1. (C). Explain **any three** of the following terms in one or two sentences:

(3)

- (i) Packing ratio (ii) A-tracts (iii) Operon (iv) satellite DNA
(v) Linking number

Q2. (A) What are the major types of DNA damage repaired by each of the following pathways:

- (a) photoreversal, (b) base excision repair, (c) nucleotide excision repair, and (d) mismatch repair
(e) SOS repair

(5)

OR

Q2. (A) Explain replication mechanism in prokaryotes

(5)

Q2. (B). What are the Okazaki fragments?

(3)

OR

Q2. (B). Draw the structure of pre initiation complex in eukaryotic replication

(3)

Q2. (C). Describe the structure of eukaryotic telomere

(2)

OR

Q2. (C). What is homologous recombination and when does it occur?

(2)

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Section II

Q3 (A). What are transcriptional activators? Describe in brief. (4)

OR

Q3 (A). Write a short note on: (4)

- i. Repression of transcription
- ii. Various basal transcription factors

Q3 (B). What are Catalytic RNA. Describe in brief (4)

OR

Q3 (B). Write a short note on: (4)

- i. Termination of transcription in Prokaryotes
- ii. Characteristics of Eukaryotic Promoters

Q3 (C). Write a note on **Any One:** (2)

- i. Bacterial RNA polymerase
- ii. Export of mRNA

Q4 (A). Write a note on **Any Two** (6)

- i. genetic code
- ii. Initiation of translation in prokaryotes Post translational modifications
- iii. Post translational modifications

Q4 (B). Draw a neatly labeled structure of **Any Two** of the following: (4)

- i. t-RNA
- ii. 43S initiation complex
- iii. Aminoacyl-AMP

Q5 (A). Attempt **Any Two** of the following. (6)

- i. Describe the regulation of *lac* operon in detail.
- ii. Explain the activation of *HO* locus in yeast.
- iii. Describe the process of chromatin remodeling.

Q5 (B). Define the following terms giving examples: (4)

- i. Operon
- ii. Repressor protein
- iii. Promoter
- iv. Trans-acting elements